

The Unique Local Counterterm between a Hénon Reflection Packet and the Full Lind Zeta

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Abstract

The orbit-resolved odd reflection-packet Euler product for the full Hénon horseshoe and the Lind zeta of the full two-shift reverse action share the same positive entropy boundary, but they do not have the same singular ledger. Starting from the primary-source formula

$$\zeta_{\text{flip}}(t) = (1 - 2t^2)^{-1/2} \exp\left(\frac{2t + 3t^2}{1 - 2t^2}\right),$$

we set $u = 1 - \sqrt{2}t$ and compare it to the packet product. The packet accounts for exponential pole coefficient $1/\sqrt{2}$, whereas the full Lind zeta has coefficient $1/\sqrt{2} + 3/4$ and a logarithmic branch of coefficient $-1/2$. We prove that

$$u^{1/2} e^{-3/(4u)} \frac{\zeta_{\text{flip}}(t)}{\mathcal{Z}_{\text{orb}}(t, 1)}$$

extends holomorphically and nonvanishingly across $u = 0$ as a local branch germ. Among counterterms $u^\beta e^{-c/u}$, the pair $(c, \beta) = (3/4, 1/2)$ is unique. Thus odd packet data admits an exact local reconciliation with the source zeta but does not by itself encode the full infinite-dihedral subgroup ledger. Global continuation, zeros, arithmetic semantics, and operator ownership remain open.

1 Two objects at one boundary

For the full n -shift with reverse flip, Kim et al. [2, Example 4.3] prove

$$\zeta_{\sigma, \rho}(t) = \frac{1}{\sqrt{1 - nt^2}} \exp\left(\frac{nt + (n + n^2)t^2/2}{1 - nt^2}\right). \quad (1)$$

This is the Lind zeta of the full infinite-dihedral action; its definition sums fixed points over all finite-index subgroups. Finite-order reversal and matrix extensions are developed in Ryu [3].

We specialize (1) to $n = 2$:

$$\zeta_{\text{flip}}(t) = (1 - 2t^2)^{-1/2} \exp\left(\frac{2t + 3t^2}{1 - 2t^2}\right). \quad (2)$$

The Hénon/full-shift conjugacy [1] also yields a different object: the product $\mathcal{Z}_{\text{orb}}(t, 1)$ over primitive odd marked reflection words. Its exact local theorem is

$$\log \mathcal{Z}_{\text{orb}}(t, 1) = \frac{1}{\sqrt{2}(1 - \sqrt{2}t)} + G_{\text{orb}}(t), \quad (3)$$

with G_{orb} analytic near $t = 1/\sqrt{2}$. Equation (3) does not assert equality with (2).

2 The source Lind boundary ledger

Put

$$u = 1 - \sqrt{2}t, \quad t = \frac{1-u}{\sqrt{2}}. \quad (4)$$

Then

$$1 - 2t^2 = u(2 - u), \quad (5)$$

$$2t + 3t^2 = \sqrt{2}(1 - u) + \frac{3}{2}(1 - 2u + u^2). \quad (6)$$

Proposition 2.1 (Full Lind local form). *There is a function G_L , analytic at $u = 0$, such that*

$$\log \zeta_{\text{flip}}(t) = \frac{1/\sqrt{2} + 3/4}{u} - \frac{1}{2} \log u + G_L(u). \quad (7)$$

Proof. The exponential term in (2) is the quotient of (6) by (5). Its residue in the coordinate u is

$$\frac{\sqrt{2} + 3/2}{2} = \frac{1}{\sqrt{2}} + \frac{3}{4}.$$

Subtracting this multiple of u^{-1} leaves a rational function analytic at zero. The prefactor contributes $-\frac{1}{2} \log u - \frac{1}{2} \log(2 - u)$, and the latter term is analytic at zero. $\square$

Remark 2.2 (Subgroup information is visible). *The coefficient $1/\sqrt{2}$ in (7) agrees exactly with the odd packet coefficient in (3). The remaining $3/4$ and the square-root branch cannot be absorbed into an analytic nonzero factor.*

3 Relative counterterm and uniqueness

Subtracting (3) from (7) gives

$$\log \frac{\zeta_{\text{flip}}(t)}{\mathcal{Z}_{\text{orb}}(t, 1)} = \frac{3}{4u} - \frac{1}{2} \log u + G_{\text{rel}}(u), \quad (8)$$

where G_{rel} is analytic.

Theorem 3.1 (Unique local counterterm). *On any fixed local branch of $u^{1/2}$,*

$$C_{\text{rel}}(t) := u^{1/2} e^{-3/(4u)} \frac{\zeta_{\text{flip}}(t)}{\mathcal{Z}_{\text{orb}}(t, 1)} \quad (9)$$

extends holomorphically and nonvanishingly across $u = 0$. Moreover, among all counterterms $u^\beta e^{-c/u}$, a nonzero holomorphic extension is possible only for

$$c = \frac{3}{4}, \quad \beta = \frac{1}{2}. \quad (10)$$

Proof. Taking a logarithm in (9) cancels the two singular terms in (8), leaving G_{rel} . Exponentiation gives the nonzero extension. For a general pair (c, β) , the logarithm contains $(3/4 - c)u^{-1} + (\beta - 1/2) \log u + G_{\text{rel}}$. A nonzero holomorphic germ has neither an essential exponential nor a nonintegral/logarithmic branch, forcing (10). $\square$

The theorem is local: the square-root branch is fixed only in a neighborhood, and no claim is made that (9) is a globally single-valued determinant.

4 Exact audit and route boundary

The accompanying certificate performs the expansion in $\mathbb{Q}(\sqrt{2})$, verifies the normalized algebraic factor $u^{1/2}/\sqrt{u(2-u)} = (2-u)^{-1/2}$, and records the analytic exponent through order six. An independent rational audit recovers $3/4$ and $1/2$. Eight tests pass in normal and optimized modes, four dependencies are locked, and 23 claim mutations are rejected.

P71 supplies a source-native local bridge: the restricted Hénon packet can be reconciled with the full Lind germ only after adding precisely the missing subgroup and branch ledgers. It does not prove global continuation, a Fredholm realization, rational-prime weights, a trace formula, or a self-adjoint operator. Route A remains exploratory and Route B is not authorized. The next test is whether the normalized relative Taylor coefficients satisfy a finite-state recurrence or carry a new obstruction.

References

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- [2] Young-One Kim, Jungseob Lee, and Kyewon K. Park. A zeta function for flip systems. *Pacific Journal of Mathematics*, 209(2):289–301, 2003. doi: 10.2140/pjm.2003.209.289.
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